

EXAMINATION INFORMATION PAGE

Written examination

Subject code: IIA4117		Subject name: Model Predictive Control	
Examination date: 21.11.2024	Examination time from/to: 09:00 – 12:00	Total hours: 3	
Responsible subject teacher: Roshan Sharma			
Campus: Porsgrunn		Faculty: Faculty of Technology	
No. of assignments: 3	No. of attachments: 0	No. of pages incl. front page and attachments: 3	
Permitted aids: None, only pen or pencil			
Information regarding attachments: No attachments			
Comments: Please use ball pen. This makes it easier for the examiner to read your answers. This exam counts 60% for the final grade.			

Select the type of examination paper

Spreadsheets

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Line sheets

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This exam counts 60% for the final grade.

Problem 1

MPC can also be used as an optimal controller to control a process where multiple objectives have to be taken into consideration simultaneously. Let us assume that you are working with a process where there are three objectives functions that are conflicting to each other.

Let us denote the three objectives as $F_1(x)$, $F_2(x)$ and $F_3(x)$. Here x are the decision variables. Assume that objective function $F_1(x)$ is the most important objective function, $F_2(x)$ is the second most important objective, and $F_3(x)$ is the least important objective function among these three conflicting objective functions.

Tasks:

- i) [15%] How would you use the hierarchical method to solve this multi objective control problem? Explain and show mathematical formulation.
- ii) [5%] Explain briefly a potential drawback or disadvantage of this method.

Problem 2

With the Kronecker product formulation of an LQ optimal control problem, the size of the control problem can easily become very large. The size of the control problem can be reduced by elimination of the equality constraints.

Let us consider an LQ optimal control problem expressed in the standard QP form as,

$$\left. \begin{array}{l} \min_z \quad \frac{1}{2} z^T H z + c^T z \\ \text{subject to} \\ A_e z = b_e \\ A_i z \leq b_i \\ z_L \leq z \leq z_H \end{array} \right\} \quad (1)$$

The problem given by equation (1) can be converted into a QP problem of the reduced form as,

$$\left. \begin{array}{l} \min_{z_2} \quad \frac{1}{2} z_2^T \tilde{H} z_2 + \tilde{c}^T z_2 \\ \tilde{A}_i z_2 \leq \tilde{b}_i \end{array} \right\} \quad (2)$$

Tasks

- (i) [15%] Explain and show in detail how QR factorization can be used to eliminate the equality constraints in order to reduce the size of the optimal control problem. Include all necessary calculations and equations.
- (ii) [5%] What are the advantages and the disadvantages of reducing the size of an optimal control problem.

Problem 3

The design of a standard MPC formulation cannot guarantee feasibility, i.e. it cannot be guaranteed that a feasible solution always exists. However, problems caused by infeasibility can be improved. A mathematical formulation of a linear MPC problem with output constraints is shown below,

$$\min_z \quad \frac{1}{2} z^T H z + c^T z, \quad (3)$$

s. t

$$A_\varepsilon z = b_\varepsilon, \quad (4)$$

$$y_L \leq y \leq y_H \quad (5)$$

where z is a vector that contains the decision variables. Equation (3) is the objective function, equation (4) is the linear equality constraints and equation (5) is the output constraints or bounds on the outputs.

Tasks:

- i) (15%) Explain in details how you can make use of slack variables to improve the feasibility of the control problem. Show also the mathematical formulation of the relaxed problem.
- ii) (5%) Explain in your own words concept about hard and soft constraints. Give some examples of each.

Roshan Sharma

12th November 2024, Porsgrunn, Norway